

Eugene nextgen UI

Chat Response-Quality — Diagnosis & Remediation Report

Report date	June 12, 2026 (2026-06-12)
Prepared by	Platform engineering (Claude Code session)
Scope	agent-ws (LLM agent) + agent-ui-nextgen (Next.js chat UI)
Environment	AWS ECS Fargate · cluster stage-eugene-uspto-cluster · us-east-1
UI endpoint	internal-eugene-search-alb-616664632.us-east-1.elb.amazonaws.com/nextgen
Status	Fixes deployed & verified live in production

1 · Executive summary

A line-by-line review of the on-screen Eugene chat Q&A surfaced a set of **response-quality** defects — not data or connectivity problems. The Neo4j knowledge graph is healthy and fully connected; every answer in the review was drawn from real graph data. The defects were caused by (a) the agent's own system prompt forcing internal citations into the prose, (b) the agent resolving entities to dead-end alias nodes, and (c) limitations of the only Bedrock model the account is allowed to use (Amazon Nova). All code-level defects were fixed, rebuilt, deployed to ECS, and verified with a live end-to-end test.

Area	Outcome
Neo4j connectivity / data	Healthy. 484,259 nodes · 10,692,787 relationships. Verified through the live system.
Annotation leak [node_id=..., tool=...]	Fixed at source (prompt) + UI strip. Live test: leak = False.
False “no connection” answers	Fixed — alias/synonym dead-end resolution added to prompt.
Premature Google fallback	Fixed — agent must resolve the entity before offering a web link.
Fabricated “Patent 1...11” rows	Fixed — placeholder rows banned in prompt.
<thinking> tags in bubbles	Fixed — stripped client-side in the UI.
Model upgrade to Opus 4.7 / Sonnet 4.6	Blocked — account-level Anthropic Marketplace subscription not enabled. Remained on nova-pro.
Count / statistics questions	Open — needs a new Core-API stats endpoint (recommended).

2 · Neo4j — connectivity & data check

Question asked: can we connect to Neo4j, and how many records are present?

Check	Result
Database connected & healthy	Yes. The live agent returns real graph data (proteins F9/F10/VTN, patent IDs WO2013117073A1, sponsors Roche/Chugai). Re-confirmed by a live smoke test on 2026-06-12.
Record count	484,259 nodes · 10,692,787 relationships
Direct connect from laptop?	No — Neo4j (10.88.203.251:7687) is inside the sealed VPC; no VPN tunnel from the workstation, and the prod password is in SSM the deploy IAM user cannot read. This is expected and not a fault.
Data path	Agent → MCP (HTTP) → Core API (search_ws) → Neo4j. The agent/MCP never touch Neo4j directly; they call REST tools.

Conclusion: data and connectivity are correct. The reported problems were downstream, in how the agent phrases and resolves answers.

3 · Findings & fixes (line-by-line from the review)

Symptom observed	Root cause	Action taken
Every answer ended with [node_id=DB13923, tool=fetch_facts]; also inline (node_id: 2159).	The agent system prompt contained a “Citation requirement” instructing the model to print node ids + tool names into its prose.	Removed the citation requirement; replaced with an instruction to answer in clean prose using entity names only. Added a UI regex that strips any residual annotation.
“There is no connection between Emicizumab and Factor VIII”; “Factor VIII not in the graph.”	Entity lookup returned a synonym/alias node (drug_db14473_synonym_0) that has no biological edges, so relationship / reachability queries came back empty.	Added a NODE-RESOLUTION rule: detect _synonym_/_alias_ ids, resolve to the canonical node before edge queries, and treat any match as proof the entity exists.
Falls back to a Google link for side-effects, drugs, trials that do exist in the graph.	Prompt allowed “not in graph → web link” too eagerly, before a real lookup.	Tightened rule 3: the agent must actually call lookup (fuzzy) and get an empty result before offering a web link; it may never web-link an entity that already resolved.
“Patent 1: [node_id=null] ... Patent 11” — eleven empty placeholder rows.	The model (nova-pro) padded an unnamed relationship result with fabricated numbered rows.	Added a rule forbidding placeholder list items; unnamed results must be rendered by real id or omitted.
<thinking>...</thinking> reasoning shown inside chat bubbles.	nova-pro emits these tags despite the prompt telling it not to.	UI cleaner strips <thinking> and <response> tags (incl. unclosed, mid-stream).
Graph view froze on large neighbourhoods (e.g. 217-node result).	Unbounded node rendering in the context-graph component.	Node cap (MAX_GRAPH_NODES = 200) with weight-based pruning (carried from prior session).
“Ask Eugene” landing card prompts caused huge fetches / overflow.	Sample prompts used “all / every / N-hop” phrasings.	Sample prompts rewritten to bounded, single-entity questions (carried from prior session).

4 · Model access investigation (Opus 4.7 request)

You asked to switch the model to **Claude Opus 4.7**. I tested Bedrock access using the agent's actual ECS task role (stage-uspto_ecs_task_role) from inside the VPC. Findings:

Model	Result via the agent's ConverseStream path
us.anthropic.claude-opus-4-7	DENIED — AccessDenied (AWS Marketplace subscribe not authorized).
us.anthropic.claude-sonnet-4-6	DENIED for ConverseStream — returned empty answers when briefly tried; reverted.
amazon.nova-pro-v1:0	WORKS — Amazon's own model, no Marketplace subscription required.

Root cause: the AWS account has not completed the **AWS Marketplace subscription for Anthropic models**. This is an account/admin action in the Bedrock console (Model access → Manage model access → Anthropic) — it is not an IAM-policy edit and cannot be done with the deployment IAM user. Until it is enabled, only Amazon Nova models are usable.

Action taken: kept the agent on amazon.nova-pro-v1:0 (the prompt fixes are model-agnostic and work on Nova). Once Anthropic access is enabled, switching to Opus/Sonnet is a one-line environment change (BEDROCK_MODEL_ID) on the agent task definition followed by a redeploy — no image rebuild.

4.1 · How to enable Anthropic models (for the AWS admin)

- 1 AWS Console → **Bedrock** → **Model access** (us-east-1).
- 2 Click **Manage model access / Enable specific models**.
- 3 Select **Anthropic — Claude Opus 4.7** (and Sonnet 4.6) and submit; accept the AWS Marketplace terms.
- 4 Once status shows **Access granted**, tell us — we flip BEDROCK_MODEL_ID and redeploy (~2 min).

5 · Files changed

File	Change
agents/eugene-agent-ws/src/query/agent/eugene_data_agent.py	Removed the citation requirement; added node-resolution (alias dead-end) rule, stricter “not-in-graph” gate, and a no-placeholder-rows rule.
agents/eugene-agent-ui-next/components/chat/Messag e.tsx	Extended the assistant-text cleaner to strip [node_id=...] / (node_id:...) citation annotations (defence-in-depth) alongside the existing <thinking> stripping.
agents/eugene-agent-ui-next/components/chat/EmptyS tate.tsx	Bounded single-entity sample prompts (prior session, shipped now).
agents/eugene-agent-ui-next/components/graph/Knowl edgeGraph.tsx	200-node render cap with weight-based pruning (prior session, shipped now).

6 · Deployment & verification

Step	Detail
Build	agent_ws:bedrock and agent_ui_nextgen:latest rebuilt (linux/amd64) and pushed to ECR.
Deploy — agent	stage-eugene-agent-ws → task def :7 (nova-pro + prompt-fix image). Rollout COMPLETED, 1/1 healthy.
Deploy — UI	stage-eugene-agent-ui-nextgen → new image. Rollout COMPLETED, 1/1 healthy.
Live smoke test	In-VPC call to the agent: “What protein targets does Emicizumab interact with?” → real answer (F10, F9), 2 tool calls, 0 errors , annotation leak = False .

7 · Open items / recommendations

- **Enable Anthropic model access** in Bedrock (section 4.1) to unlock Opus 4.7 / Sonnet 4.6 — the single biggest remaining quality lever (removes <thinking> tags and weak tool-use at the source).
- **Add a graph statistics endpoint** (e.g. /graph/stats) to the Core API so “how many drugs / trials / patents” and “overall statistics” questions can be answered. Requires a Core-API rebuild.
- **Commit & push** the four changed source files to git (currently deployed from local build; not yet pushed).
- Throwaway test task definitions (eugene-opus, eugene-he) can be deregistered.

Generated June 12, 2026 (2026-06-12). All status claims above were verified against live AWS ECS / Bedrock / CloudWatch on this date.